1 Solution — GIAM 9.2 Problem 2

1.1 Problem

"Escape of the clones" is a nice puzzle, originally proposed by Maxim Kontsevich. The game is played on an infinite checkerboard restricted to the first quadrant — squares are identified with points (x, y) with $x, y \in \mathbb{Z}_{>0}$. A *clone move* is legal when the cells immediately above and to the right of a clone are *both empty*; the clone then moves one cell up while a copy is placed in the cell to the right.

We begin with three clones occupying $(1, 1)$, $(2, 1)$, and $(1, 2)$ — call those three cells the **prison**. Can the three clones escape the prison?

1.2 Answer

No — escape is impossible.

The proof copies the Section 9.2 strategy: invent a quantity (an *invariant*) that the moves preserve, then show that any "escaped" configuration violates the conservation law.

1.3 Setup — A Move Replaces One Clone by Two

A clone move at cell (x, y) has the effect:

$$\{(x, y)\} \quad \longmapsto \quad \{(x, y + 1),\ (x + 1, y)\},$$

provided both $(x, y + 1)$ and $(x + 1, y)$ were empty before the move. The single occupied cell (x, y) becomes empty; the two cells above and to the right become occupied.

The number of clones grows by exactly 1 per move. After n moves, there are $3 + n$ clones at $3 + n$ *distinct* cells — always a *finite* configuration.

1.4 The Invariant — Weight Each Cell by $1/2^{x+y}$

To each first-quadrant cell, assign the weight

$$f(x,y) \;=\; \frac{1}{2^{x+y}} \;=\; \left(\tfrac{1}{2}\right)^{x+y}.$$

The **value of a configuration** is the sum of weights over occupied cells:

$$V \;=\; \sum_{(x,y)\,\in\,\text{occupied}} f(x,y).$$

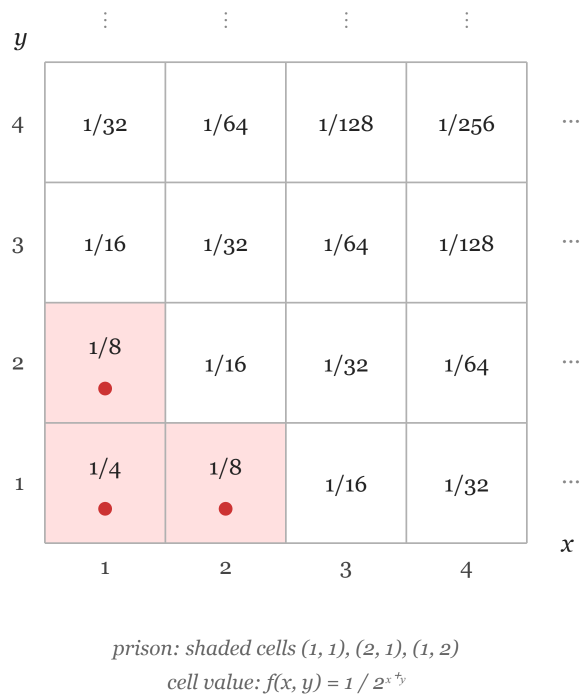

Figure 1.1: First-quadrant cells labeled with their weight $1/2^{x+y}$. The three shaded cells (1, 1), (2, 1), (1, 2) — the prison — are initially occupied by clones.

1.5 Invariance Under Clone Moves

Claim. Every legal clone move preserves V.

Proof. The move at (x, y) removes the contribution $f(x, y)$ and adds $f(x, y+1) + f(x+1, y)$. So

$$\Delta V \;=\; f(x, y+1) \;+\; f(x+1, y) \;-\; f(x, y).$$

Substituting f:

$$\begin{aligned} \Delta V \;&=\; \tfrac{1}{2^{x+y+1}} \;+\; \tfrac{1}{2^{x+y+1}} \;-\; \tfrac{1}{2^{x+y}} \\ &=\; \tfrac{2}{2^{x+y+1}} \;-\; \tfrac{1}{2^{x+y}} \\ &=\; \tfrac{1}{2^{x+y}} \;-\; \tfrac{1}{2^{x+y}} \;=\; 0. \qquad \blacksquare \end{aligned}$$

The choice $f(x, y) = 1/2^{x+y}$ was *engineered* for this: the recurrence $f(x, y) = f(x, y+1) + f(x+1, y)$ forces f to halve under each unit step, hence the base $\frac{1}{2}$.

1.6 Initial Value V_0

The starting configuration has clones at the three prison cells:

$$V_0 \;=\; f(1,1) \;+\; f(2,1) \;+\; f(1,2) \;=\; \tfrac{1}{4} \;+\; \tfrac{1}{8} \;+\; \tfrac{1}{8} \;=\; \tfrac{1}{2}.$$

By the invariance claim, *every reachable configuration also has value* $\frac{1}{2}$.

1.7 The Total Weight of the First Quadrant is 1

Sum the weights over *all* cells of the first quadrant:

$$\sum_{x \ge 1} \sum_{y \ge 1} \tfrac{1}{2^{x+y}} \;=\; \left(\sum_{x \ge 1} \tfrac{1}{2^x} \right) \cdot \left(\sum_{y \ge 1} \tfrac{1}{2^y} \right) \;=\; 1 \cdot 1 \;=\; 1.$$

Each row-sum and column-sum is the geometric series $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$. The double sum splits since $2^{x+y} = 2^x \cdot 2^y$.

Therefore the **non-prison** cells have total weight

$$\sum_{(x,y) \notin \text{prison}} f(x, y) \;=\; 1 \;-\; V_0 \;=\; 1 \;-\; \tfrac{1}{2} \;=\; \tfrac{1}{2}.$$

1.8 The Contradiction

Suppose, toward contradiction, that the clones can escape — i.e. there is a sequence of legal moves ending in a configuration with *no* clone in the prison.

Let E be the set of cells occupied at the escape moment. By the setup:

- E is a **finite** subset of the first quadrant (after finitely many moves, we have finitely many clones).
- $E \cap \textsf{prison} = \emptyset$ (this is what "escape" means).
- $V_E = V_0 = \frac{1}{2}$ (invariance).

But then

$$\tfrac{1}{2} \;=\; V_E \;=\; \sum_{(x,y)\in E} f(x,y) \;\le\; \sum_{(x,y)\,\notin\,\text{prison}} f(x,y) \;=\; \tfrac{1}{2}.$$

So the inequality is forced to be an *equality* — E must cover *every* non-prison cell. That is impossible: E is finite and there are *infinitely many* non-prison cells, each with strictly positive weight. The inequality is therefore *strict*:

$$V_E \;<\; \tfrac{1}{2}.$$

This contradicts $V_E = \frac{1}{2}$. So no escape sequence can exist. ■

1.9 A Sanity Check — Verifying V on a Single Move

Start: clones at $(1,1),(2,1),(1,2)$, total $V = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$.

The clone at $(2,1)$ has empty neighbours at $(2,2)$ and $(3,1)$, so its move is legal. After the move:

- Cells occupied: $(1,1),\ (1,2),\ (2,2),\ (3,1)$.
- New value: $\frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$. ✓

The invariant survives. The clone at $(1,2)$ is symmetric and gives the same check. The clone at $(1,1)$ *cannot* move from the initial configuration, since its right neighbour $(2,1)$ and upper neighbour $(1,2)$ are both occupied.

1.10 Remark — How This Compares to the Section 9.2 Invariant

The Section 9.2 argument and this one share a single template:

step	Section 9.2 (checker jumps)	Kontsevich clones
Choose a weight f such that the move equation...	$f(p) \geq f(p_{\text{mid}}) + f(p_{\text{land}})$ (inequality)	$f(p) = f(p_{\text{up}}) + f(p_{\text{right}})$ (equality)
...forces a parameter to satisfy...	$r^2 + r = 1$ (golden ratio)	base $= \frac{1}{2}$
Compute initial total V_0	1 (army on $y \leq 0$)	$\frac{1}{2}$ (prison)
Compute total over the “winning” region	”≥ 1” required to reach $(0, 5)$	$\frac{1}{2}$ (non-prison)
Contradiction	Reaching $(0, 5)$ would require strictly more than V_0 from a finite config	Escape would require occupying *every* non-prison cell — impossible for a finite config

Table 1.1

Two differences worth flagging:

- **Equality vs. inequality.** In 9.2 the move *can* decrease V (jumps away from the target lose mass); in the clone puzzle every legal move conserves V exactly. Both versions still produce monotonicity: V never *increases*.
- **Where finiteness bites.** In 9.2 the contradiction uses that a finite configuration of checkers cannot cover the entire half-plane needed to keep $V = 1$. Here it uses that a finite set of cells cannot exhaust the non-prison weight $\frac{1}{2}$.

The shared trick: an *exponential* weight, geometric series for the totals, finiteness of any reachable configuration.

1.11 Remark — Why 1/2 and Nothing Else

The move equation $f(x,y) = f(x,y+1) + f(x+1,y)$ is what fixed the choice of f. Trying $f(x,y) = c^{x+y}$ for some unknown $c > 0$:

$$c^{x+y} \;=\; c^{x+y+1} \;+\; c^{x+y+1} \;=\; 2\,c^{x+y+1},$$

so $1 = 2c$, forcing $c = \frac{1}{2}$. Any other exponential rate either *fails* the invariance (so V wouldn't be conserved) or — if $c \geq 1$ — fails to give a finite total over the first quadrant.

The same construction generalises: change the move rule, change the recurrence, and the recurrence selects a new base. The clone puzzle has the cleanest possible such selection — a single arithmetic identity $1 = 2 \cdot \frac{1}{2}$.

1.12 Remark — Why the Strict-Inequality Step is Essential

The conservation law $V = \frac{1}{2}$ and the bound $V_{\text{non-prison}} = \frac{1}{2}$ together do *not* immediately rule out escape — they only show that the escape configuration would *saturate* the bound. The contradiction comes from the additional fact that the saturation is unachievable by any *finite* configuration:

$$\text{finite } E \subseteq \text{non-prison} \implies \sum_{(x,y)\in E} f(x,y) \;<\; \tfrac{1}{2}.$$

This is true because every non-prison cell has *strictly positive* weight, and the non-prison region has infinitely many cells. Omitting any cell strictly decreases the sum.

If we'd allowed *infinite* configurations the argument would collapse: an infinite army filling the non-prison region would have $V = \frac{1}{2}$ and could in principle have "escaped". The finiteness of the clone count — three to begin with, $3 + n$ after n moves — is what closes the trap.

1.13 Remark — What If We Allowed the Full Plane?

Suppose we *removed* the first-quadrant restriction and let clones move into negative-coordinate cells too. The weight function $f(x,y) = 1/2^{x+y}$ would still satisfy the move equation, but the total mass

$$\sum_{x,y\in\mathbb{Z}} \frac{1}{2^{x+y}}$$

is *infinite* (each term $1/2^{x+y}$ becomes arbitrarily large for negative $x+y$). The book-keeping that powered our contradiction — "the non-prison region has total weight $\frac{1}{2}$" — evaporates.

So the first-quadrant restriction is doing real work. It isn't a cosmetic constraint; it is the geometric ingredient that makes the geometric-series totals *finite*, hence the invariant argument actually pin something down.

1.14 Remark — The Clones Are "Almost" Reproducing

Each move adds one clone. After n moves there are $3+n$ clones. Yet V stays fixed at $\frac{1}{2}$. How can the population grow without the total mass increasing? Because the cells they occupy must drift further from the origin — and weights $1/2^{x+y}$ shrink exponentially in $x+y$.

Intuitively: the clones can spread out to arbitrarily many cells, but only by occupying cells whose individual weights are tiny. The total weight is conserved by the geometric trade: each split replaces one cell of weight w with two of weight $w/2$ each.

This is also why the proof works pessimistically: even if every move spreads clones out into as many cells as possible, the *total* contribution stays the same. There is no "smart" sequence of moves that gains mass.